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Homework 6

1.

Exercise 6.7 Referring to Example 2.7, consider a $\mathcal{B}e(2.7, 6.3)$ target density.

a. Generate Metropolis–Hastings samples from this density using a range of independent beta candidates from a $\mathcal{B}e(1, 1)$ to a beta distribution with small variance. (*Note:* Recall that the variance is $ab/(a+b)^2(a+b+1)$.) Compare the acceptance rates of the algorithms.
b. Suppose that we want to generate a *truncated* beta $\mathcal{B}e(2.7, 6.3)$ restricted to the interval (c, d) with $c, d \in (0, 1)$. Compare the performance of a Metropolis–Hastings algorithm based on a $\mathcal{B}e(2, 6)$ proposal with one based on a $\mathcal{U}(c, d)$ proposal. Take $c = .1, .25$ and $d = .9, .75$.

(a)

(b)

2.

Exercise 6.9 Calculate the mean of a gamma $\mathcal{G}(4.3, 6.2)$ random variable using

a. Accept–Reject with a gamma $\mathcal{G}(4, 7)$ candidate;
b. Metropolis–Hastings with a gamma $\mathcal{G}(4, 7)$ candidate;
c. Metropolis–Hastings with a gamma $\mathcal{G}(5, 6)$ candidate.

In each case, monitor the convergence across iterations.

(a)

(b)

(c)

3.

4.